BRANDON SANTOR

GEOSPATIAL RESEARCH | SCIENTIFIC SOFTWARE | U.S. MARINE CORPS VETERAN

brandon.santor@tamu.edu | emp1241.github.io

Geospatial researcher and software developer building scientific tools from data preparation through the interface. My work spans tree-growth and climate analysis, hydrothermal-vent mapping, reconstructed Martian terrain, spatial evidence systems, and independent research on smartphone satellite positioning. I study geographic information science and technology at Texas A&M and served in the U.S. Marine Corps.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant & Software Developer | Torbenson Lab

March 2026-present | Paid appointment since June 2026

- Independently built the Tree-Ring Atlas Explorer (TRAE) in response to lab needs; its research use led to a paid appointment for continued software development and research support.
- Added reviewed data imports, linked maps and graphs, climate matching, saved work sessions, and exports that record source and method details. Built tools for PRISM gridded climate data, NetCDF multidimensional files, ROXAS wood-anatomy outputs, and year-labeled WalDendro X-ray records.
- Used TRAE to derive summaries, organize study data, and prepare exports for a co-authored manuscript in preparation; its screening summaries support analysis but do not replace scientific validation.

Research Lead & GIS Developer | New World Screwworm (NWSW)

Fall 2026-present | Texas A&M GIS Practicum

- Lead a team studying where New World Screwworm has been reported. Check official records for reporting differences and possible repeated events, and build a repeatable ArcGIS process that turns checked sources into geographic data and maps.
- Tested how much random variation remained in a location-uncertainty calculation. A completed 64-million-trial Monte Carlo experiment met the study's numerical precision target; it did not establish exact detection sites.
- Independently built a separate spatial evidence application: FastAPI serves the browser interface, PostGIS links records to geographic features, and TypeScript and Leaflet support mapped review.

Independent Esri GIS Technical Consultant & Instructor

February-May 2026

- Audited seven graduate GIS modules and rebuilt six workflows into 189 pages of illustrated technical manuals.
- Corrected GIS software and projection problems, misaligned map grids, database structure and geographic-join errors, licensing issues, and desktop-to-web failures.
- Built quality-control checkpoints and taught reproducible workflows that enabled the client to execute corrected analyses independently.

High School Baseball Coach

2013-2020 | Paid program coach

- Coached nearly 100 student-athletes each year, developing baseball skills, teamwork, and discipline.
- Coordinated conditioning, equipment, and maintenance for the program.
- Mentored student-athletes and emphasized education throughout their high school experience.

U.S. Marine Corps | Infantryman, Acting Squad Leader & Logistics NCO

2011-2013

- Managed weapons, ammunition, medical supplies, and safety equipment supporting approximately 350 Marines.
- Served as acting squad leader for 13 Marines.
- Prepared Marines for combat through live-fire training.

TEKsystems | Network & Computer Systems Contractor

2007-2011

- Executed enterprise hardware and software deployments involving 100+ devices per project during nationwide infrastructure upgrades.
- Independently diagnosed complex computer and network systems issues across changing client environments.
- Installed, configured, and supported systems while coordinating with local and remote technical teams.

SELECTED SOFTWARE & RESEARCH

Mars Explorer

Team lead | Database & 3D systems

- Lead a five-person team and build the PostGIS database, processing pipelines, read-only API, and Cesium terrain system. The working preview connects four Curiosity areas with bounded orbital-data terrain, twelve curated Navcam views, and separately preserved instrument observations.

Thermal Vent Atlas Explorer

Independent creator & developer

- Built and delivered TVAE to Sylvan Lab for review and internal use, mapping 721 preserved InterRidge vent-field records with seafloor relief, tectonic boundaries, local research fields, comparisons, and saved sessions.

COLE | Cost Of Living Explorer

Application creator | Developer | Project lead

- Created and lead COLE and contributors' feature additions. Released version 1.0.1 for web and Windows so people can compare household budgets, jobs, climate, population profiles, hazards, and public outdoor space across counties.

Research Architect (RA)

Creator & principal designer | PhalAInx independent software studio

- Created Research Architect for human research teams, with individual AI assistant kits, shared sources, review, and recovery. The 2.5.1 preview records setup answers and approval before kit generation; earlier Hub improvements remain in use in NWSW.

High Command (HC)

Creator & developer | PhalAInx independent software studio

- Built a Windows workspace for projects, AI conversations, tasks, and recoverable imports using Tauri, React, Rust, and SQLite. A separate Codex pilot passed live interaction checks; newer command-stop and recovery work remains staged for testing.

The Deception Gap | Smartphone GNSS

Independent continued research

- Expanded a class project into independent satellite-positioning research, more than doubling the original collection to 781 retained readings from three smartphones across five crowd categories; led continued field collection, statistical analysis, interpretation, and poster development.

NASA SUITS Challenge Proposal

Team lead & primary system designer | 2021-2022

- Led a student team's HoloLens 2 interface proposal for extravehicular activity (EVA), connecting local positioning, terrain overlays, and a command hub intended to distribute current crew positions and shared pins; co-authored technical documentation submitted to NASA for review.

EDUCATION

Texas A&M University

Expected May 2027 | B.S. in Geographic Information Science & Technology | Institutional GPA: 4.0 / 4.0

A.A. Liberal Arts

Blinn College | 2024

A.S. Computer Networking

ITT Technical Institute | 2007

HONORS & SCHOLARSHIPS

Second Place | TxGIS Day Undergraduate Student Research Presentation Competition | 2025 | Health Geography GIS and statistical analysis

Galloway '29 Aggie Veteran Freedom Scholarship II | Two-time recipient | 2025-2027

Dennis W. Hughes '79 & Melinda Hood Hughes '81 Endowed Scholarship in Geography | Two-time recipient | 2025-2027

Richard A. & Mollie B. Williford Dean's Scholarship | College of Arts & Sciences | 2026-2027

TECHNICAL TOOLKIT

Languages: Python; JavaScript / TypeScript; HTML / CSS; Rust; SQL.

Geospatial tools: ArcGIS Pro / Field Maps; QGIS / GDAL; GeoPandas / Shapely / PyProj / Pyogrio; PostgreSQL / PostGIS; Cesium / Leaflet; Rasterio.

Application development: FastAPI / Flask / Waitress; React / Vite / Tauri 2; Node.js / Caddy; SQLite; PyInstaller / Windows packaging; JSON Schema.

Research methods & data: NumPy / pandas; NetCDF / PRISM / RWL / CRN; Monte Carlo precision checks; Smartphone satellite-positioning field analysis; Research source tracking; SHA-256 manifests.

RESEARCH COMMUNICATION

Co-authored manuscript in preparation with Torbenson Lab; scientific workflow development and research presentation.